

Exam 1 Review Day

Confusion matrices

Y

	0	1
$\hat{Y}$ 0		
1		

Accuracy, Sensitivity, Specificity, PPV

$$\hat{Y}_i = \begin{cases} 1 & \hat{r}_i > \text{threshold} \\ 0 & \text{else} \end{cases}$$

as threshold $\uparrow$: sens $\downarrow$, spec $\uparrow$

threshold of 0.5 typically maximizes accuracy,
but accuracy on its own is a bad metric for imbalanced data

Full model: (HW 5)

over Asking_i ~ Bernoulli(π_i)

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \text{Area}_i + \beta_2 \text{Age}_i + \beta_3 \text{Suburban}_i + \beta_4 \text{Urban}_i$$

Research question: is there a relationship between location & probability of paying over asking, after accounting for Area & Age?

H_0 : $\beta_3 = \beta_4 = 0$ H_A : at least one of $\beta_3, \beta_4 \neq 0$

reduced model: the model under H_0

over Asking_i ~ Bernoulli(π_i)

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 \text{Area}_i + \beta_2 \text{Age}_i$$

Test Statistic : drop-in - deviance

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

Under H_0 , $G \sim \chi^2_{\ell}$ $\ell = \# \text{ parameters tested}$

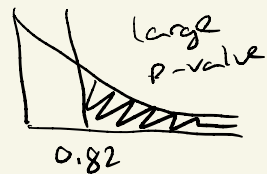
HW S: $\ell = 2$ (β_3, β_4)

$$\begin{aligned} G &= 256.45 - 255.63 \\ &= 0.82 \end{aligned}$$

p-value: Compare to a χ^2_2

R: $\text{pchisq}(0.82, 2, \text{lower.tail} = \text{FALSE})$

χ^2_2 : mean 2, variance of 4
 $\Rightarrow 0.82$ is not unusual



Effect size (in terms of standard deviations)

$$\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$\hat{\beta}_1$: change in log-odds associated with a one-unit increase in X

$e^{\hat{\beta}_1}$: change in odds (multiplicative) associated with a one-unit increase in X

$SD(X) \cdot \hat{\beta}_1$: change in log-odds associated with a one-SD increase in X

$$SD(X) \cdot \hat{\beta}_1$$

$e^{SD(X) \cdot \hat{\beta}_1}$: change in odds (multiplicative) associated with a one-SD increase in X

Likelihoods { maximum likelihood

$$\text{likelihood} = P(\text{observed data} \mid \text{model})$$

maximum likelihood: choose parameters in model that fit the observed data best (i.e., that maximize likelihood)

e.g. logistic regression:

$$Y_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 X_i$$

parameters to estimate: β_0 and β_1

1) look at $P(\text{observed } Y \text{ values} \mid \beta_0, \beta_1, X \text{ values})$

2) Choose β_0, β_1 to maximize this probability

MLE via calculus:

- 1) write down likelihood function
(depends on problem & observed data)
- 2) Take log (calculate log-likelihood)
- 3) Differentiate log-likelihood wrt parameter of interest
- 4) set derivative = 0, solve for parameter of interest

calculus skills:

- 1) Derivatives (logs, powers, etc.)
- 2) chain rule
- 3) Solve an equation

Example interpretation:

$$\log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = -3 + 0.02 GRE_i + 0.4 GPA_i$$

(made-up numbers)

an increase of a 1 point in GRE score is associated with an increase in the predicted log odds of admission by 0.02, holding GPA constant

an increase of 1 point in GRE score is associated w/ an increase in the odds of admission by a factor of $e^{0.02} \approx 1.02$, holding GPA (i.e., a 2% increase in odds) constant